

Sinusoid

Montague Hamilton

2 October 2026

1 Questions

1.

$$h(t) = c + a \sin\left(\frac{2\pi t}{p} + \phi\right)$$

For a wheel centred at height 4, radius 3, period 12; the rider starts at the top. At $t = 3$,

$$\begin{aligned} h(t) &= 4 + 3 \sin\left(\frac{2\pi 3}{12} + \frac{\pi}{2}\right) \\ &= 4 + 3 \sin(\pi) \\ &= 4 \end{aligned}$$

So at the same level as the wheels centre.

2. **Describe fully (amplitude, midline, period):** $y = 5 \sin(2x) - 1$

The midline is -1 , so the function starts at -1 and centres around it.

The amplitude is 5, so the function reaches 5 higher and 5 lower than midline. So the maximum would be 4 and the minimum would be -6 .

The period is π , so the function repeats twice for every increase in x .

3. **Convert a period of 6 seconds – what is the angular frequency? And what phase ϕ describes a rider who starts at the midline and moves downwards?**

Frequency is the inverse of the period, so

$$\begin{aligned} f &= \frac{2\pi}{p} \\ f &= \frac{\pi}{3} \end{aligned}$$

The two simplest phases that work for a rider starting at the midline and moving downwards is $\pm \pi$

4. **True or false with a reason: doubling a wheel's amplitude doubles its period?**

False, doubling the amplitude doubles the linear speed, the distance over time, but it doesn't change the angular speed, the speed of the change in angle over time, so the period stays the same.

5. **Spot the error: a student writes $h(t) = 5 + 2 \sin\left(\frac{\pi t}{4}\right)$ and claims the rider starts at the bottom. Test at $t = 0$ then fix it.**

At $t = 0$, $\sin(0) = 0$ so $h(t) = 5$. This is problematic because 5 is the midline of the circle. To offset the change in starting location, a correct phase must be used,

$$h(t) = 5 + 2 \sin\left(\frac{\pi t}{4} - \frac{\pi}{2}\right)$$